Introduction to Econometrics

Section 2A / 2B

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Instructions

For full credit, you must upload:

1. **handwritten answers to parts 1 & 2**. Please write clearly, number your answers, and put them in order.
2. **handwritten or typed answers to parts 3 & 4**. Please type or write clearly, number your answers, and put them in order.
3. **a single do-file** containing the code you used to answer **parts 3 and 4**. The do-file must run without errors and produce all of the requested output for full credit. Formatting tips:
   - At the top of the do-file, please include a comment line with your name; you should also add a comment line before the relevant code for each part of each question (e.g., 3.a).
   - Please follow the problem set submission guide and do-file template for the basic format. (A link will be posted in the Canvas assignment.)
4. **four figures** as **pdf** or **image** files, as per questions **3(c), 3(f), 4(b)** and **4(c).**

Aside from the figures, you are not required to upload Stata output (although you may copy and paste it into the document with your written answers if you find that helpful). However, the answers you report should be based on output that is produced by the Stata code you submit.

Part 3 - Hints

Part a:

- If I just loaded a dataset, what's the first command you'd run to understand what's inside?
- What command do you use to summarize the variables?
- Do you see any variables where the number of observations is smaller than others?
- How would you spot missing data quickly?

Part 3 - Hints

Part a - Answers:

- If I just loaded a dataset, what's the first command you'd run to understand what's inside? *describe*
- What command do you use to summarize the variables? *summarize*
- Do you see any variables where the number of observations is smaller than others? *hrswork_wk, educ_mom, educ_pop*
- How would you spot missing data quickly? *Missing data shows up as different observation counts across variables*

Part 3 - Hints

Part b:

- How would you translate 'experience = age education' into Stata?
- What happens if educ is missing? What will exper be?
- What condition defines full-time here?
- How do we translate 'greater than 35' into code?

Part 3 - Hints

Part b - Answers:

- experience = age education → *gen exper = age - educ*
- If educ is missing → exper will also be *missing*
- Full-time = more than 35 hours/week
  initialize an empty variable, *gen fulltime = 0*
  replace fulltime = 1 if hrswork_wk > 35
- If hrswork_wk is missing → hrswork_wk will also be *missing* - how to code?
- Why do we handle missing separately? *Logical conditions can overwrite missing values*

Part 3 - Hints

Part c:

- Which option gives us median and percentiles, not just mean and sd?
- Which command shows us a distribution?
- How to save the figure? How to save it as an image (.png, .jpeg) or pdf?

Part 3 - Hints

Part c - Answers:

- Detailed stats → *summarize exper*, **detail**
- *histogram exper, w(2)*
  If I change w(2) to w(5), what happens?
- *graph export "3c_hist_exper.png", replace*

Part 3 - Hints

Part d:

- If a variable is categorical, should we use summarize?
- How do we see frequencies of each category?

Part 3 - Hints

Part d - Answers:

- No, use *tab occup_cat*
- Use *tab occup_cat, sort* to find most frequent ones

Part 3 - Hints

Part e:

- What does "separately by group" mean in Stata?
- What are two ways to compute group averages?
- How can you compute the share of workers who are paid by the hours? Which variable denotes it?

Part 3 - Hints

Part e - Answers:

- Use *if* conditions or *tab ...*, *sum()*
- Example:
  mean wage if fulltime == 1 OR
  tab fulltime, sum(wage) - creating a frequency table for wages.
- *paid_hrly*

Part 3 - Hints

Part f:

- What command do you use for producing correlation matrix?
- What counts as a "strong" correlation?
- What command to use for a scatter plot?
- How do you submit the figure as an image or pdf?
- Does it make sense to correlate wage with industry?

Part 3 - Hints

Part f - Answers:

- correlation matrix - *corr x y z a*
- Strong $\approx$ closer to 1 in absolute value
- *scatter x y*
- *graph export "xxx.pdf", replace*
- Industry is categorical $\rightarrow$ correlation is not meaningful

Part 3 - Hints

Part g:

- How do we convert weekly hours to yearly hours?
- If we multiply a variable by 52, what happens to:
  - Mean?
  - Standard deviation?
  - Correlation?
- How do you compute the mean, standard deviation?
- how to find the correlation between two variables?

Part 3 - Hints

Part g - Answers:

- Weekly to yearly hours: *gen hrswork_yr = hrswork_wk * 52*
- Mean and SD scale by 52
  Correlation does **not** change
- Use *summarize*
- same as part f

Part 4 - Hints

Part a:

- Which command gives you a table with average rent per person by the number of roommates along with counts?
- Which estimate would you trust more: 2 roommates or 4 roommates?

Part 4 - Hints

Part a - Answers:

- Use *tab roommates, sum(rent)*
- Larger sample size $\rightarrow$ more reliable estimate

Part 4 - Hints

Part b:

- What command do you use for seeing the frequency distribution visually?
- Why do we use discrete freq?

Part 4 - Hints

Part b - Answers:

- use *histogram roommates, discrete freq*
- Run the same command without *discrete freq* to see the difference

Part 4 - Hints

Part c:

- How do you produce the first scatter plot?
- How to generate the conditional mean of rent (that is on the number of roommates)?
- Why overlay the mean on the scatter plot?

Part 4 - Hints

Part c - Answers:

- Use *scatter x y*
- *egen mean_rent = mean(rent), by(roommates)*
- Overlay shows the trend clearly - use *scatter rent mean_rent roommates*
  For more specifications on font and colors, use *help scatter*

Big Picture Takeaways

- What is one thing correlation cannot tell us?
- Why is grouping useful in data analysis?
- When should we trust a sample average?